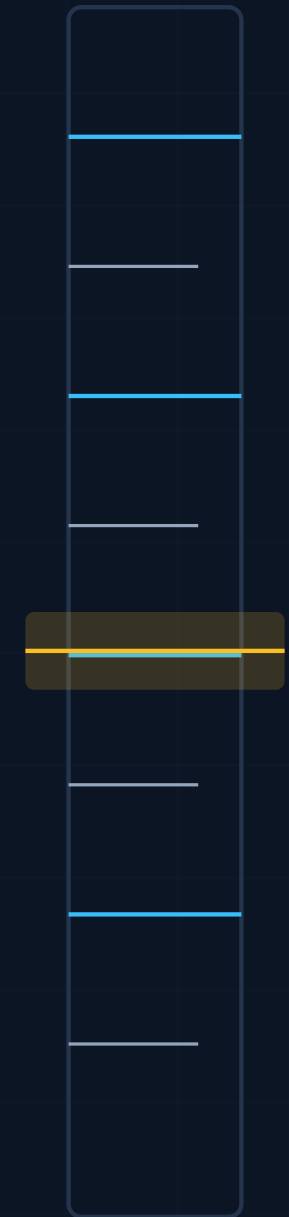


TOPIC

Measurements & Experimentation

Standards, instruments, and the discipline of accurate measurement.



— WHAT IS MEASUREMENT

Comparing a quantity with a standard

Measurement is the process of comparing an unknown physical quantity with a known fixed quantity of the same kind, called a unit.

Numerical value

How many times the unit is contained in the quantity.

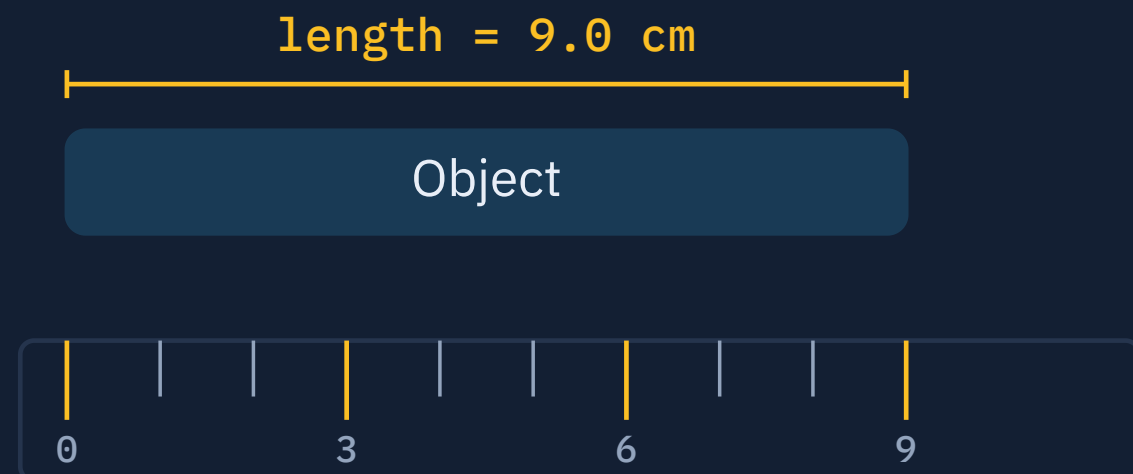
Unit

The standard chosen for comparison, e.g. metre or gram.

Standard unit

A unit agreed worldwide so results match everywhere.

MEASURING LENGTH



Reading = numerical value (9.0) + unit (cm)

— SI SYSTEM

The seven base quantities

Standard units let scientists everywhere compare results without confusion. Every other unit is built from these seven.

QUANTITY	UNIT	SYMBOL
Length	Metre	m
Mass	Kilogram	kg
Time	Second	s
Temperature	Kelvin	K
Electric current	Ampere	A
Luminous intensity	Candela	cd
Amount of substance	Mole	mol

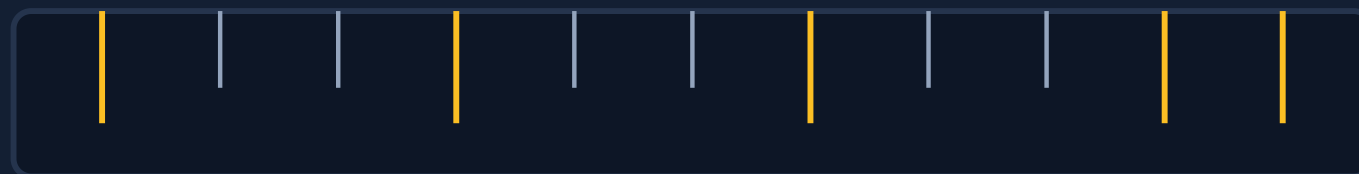
LEAST COUNT

The smallest value an instrument can read

A smaller least count means a more precise instrument and a more accurate experiment.

Standard ruler

LC = 1 mm



Each smallest division equals one millimetre.

Digital stopwatch

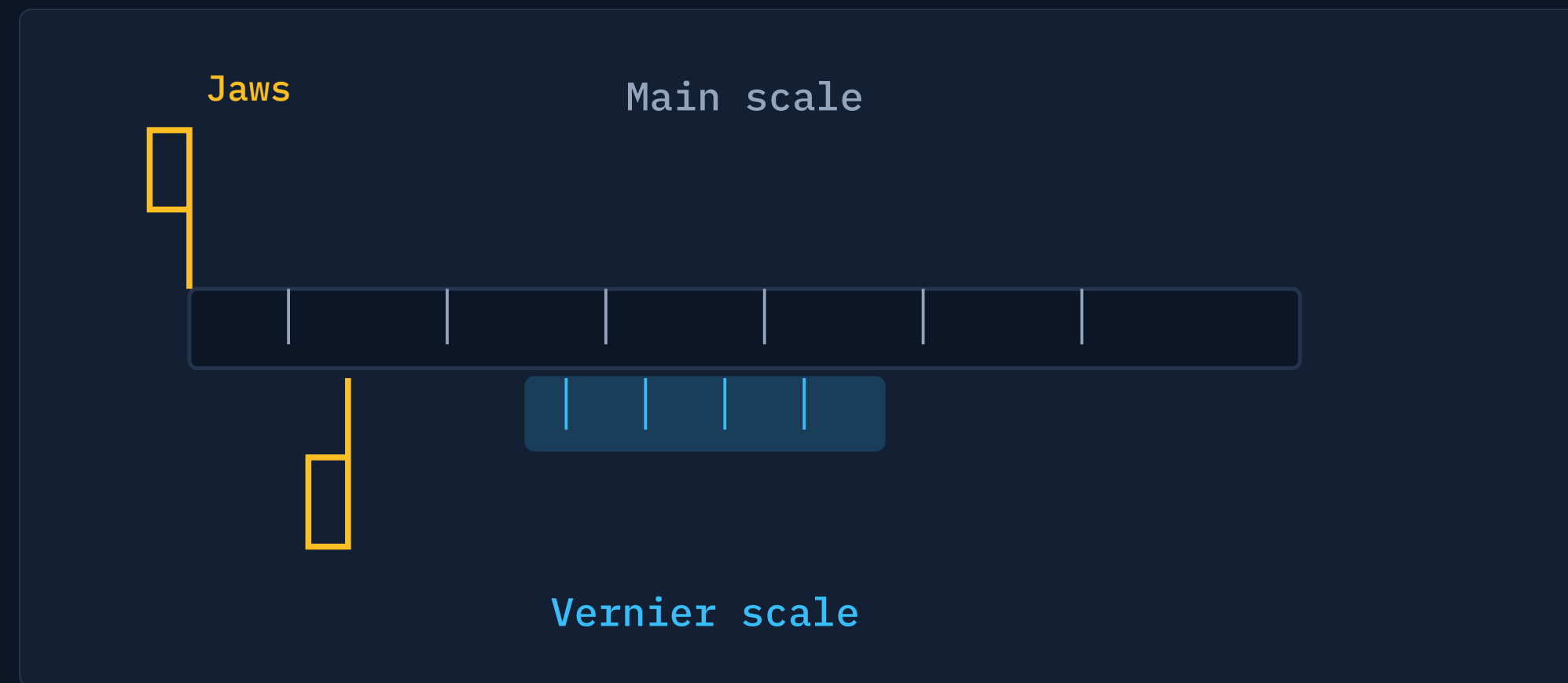
LC = 0.01 s

00:42.57

Reads to a hundredth of a second.

— VERNIER CALIPER

Reading beyond the main scale

**LEAST COUNT**

1 main scale division ÷ total vernier divisions

$$1 \text{ mm} \div 10 = 0.1 \text{ mm}$$

Main scale — gives the whole reading.

Vernier scale — adds the fine fraction.

Jaws — grip the object to be measured.

— SCREW GAUGE · MICROMETER

Measuring very small thicknesses

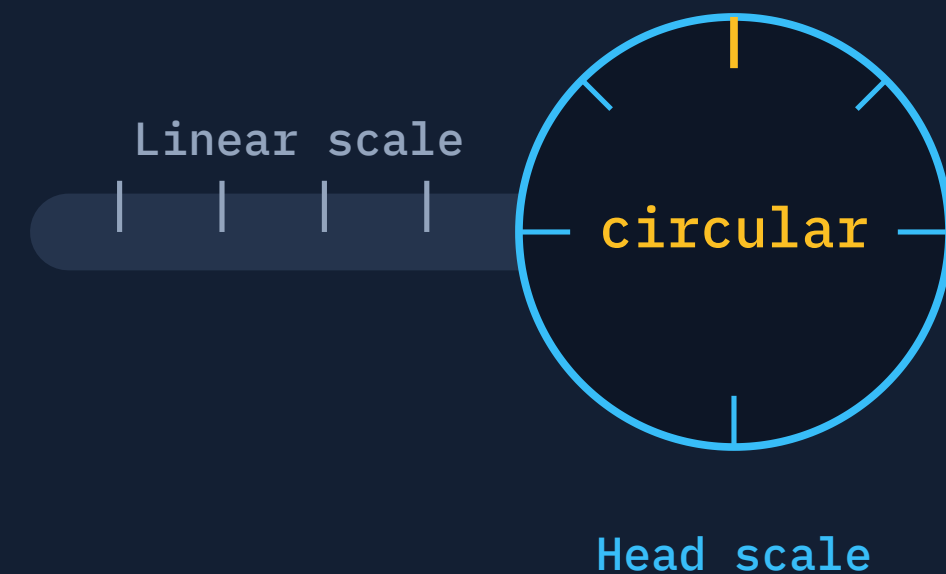
PITCH

Distance the screw advances along the linear scale in one full rotation of the head.

LEAST COUNT

$$\text{Pitch} \div \text{head scale divisions}$$

Used to measure the diameter of thin wires and the thickness of thin sheets.



— SIMPLE PENDULUM

A clock built from a swinging bob

01 Oscillation

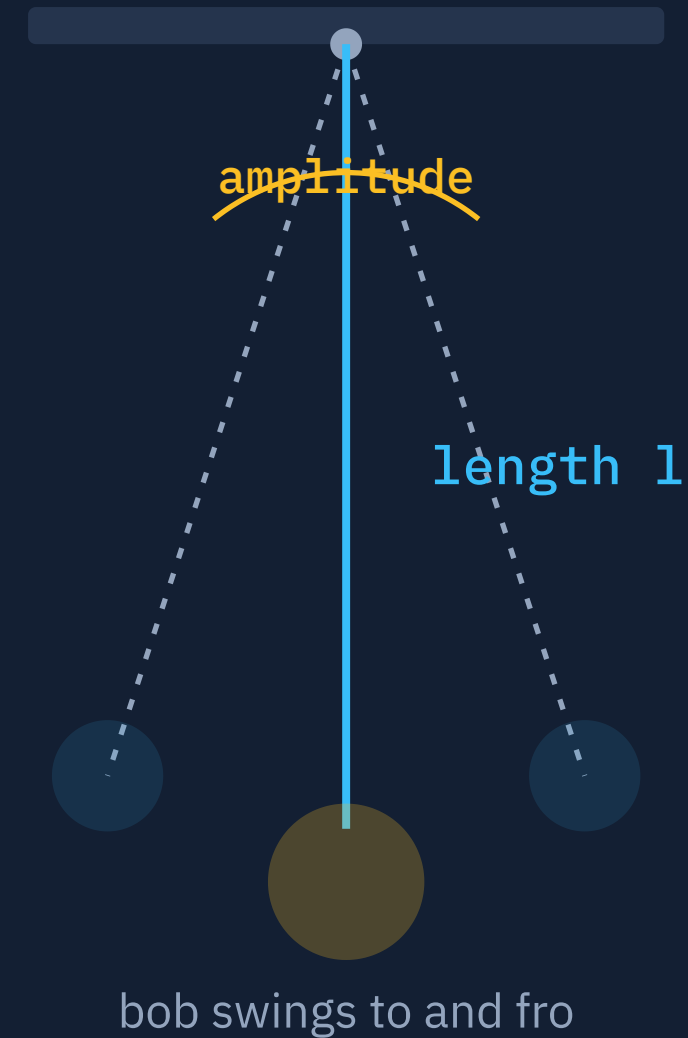
One complete to-and-fro swing of the bob.

02 Time period (T)

Time taken for one complete oscillation.

03 Frequency (f)

Number of oscillations per second, $f = 1 / T$.



WHAT AFFECTS THE TIME PERIOD

Length and gravity — not mass

$$T = 2\pi\sqrt{1 / g}$$

Length (l)

Longer pendulum → larger time period. T increases with $\sqrt{l}$.

depends ↑

Gravity (g)

Stronger gravity → smaller time period. T decreases as g rises.

depends ↓

Mass of bob

No effect at all — heavy and light bobs swing in the same time.

no effect ×

 ERRORS IN EXPERIMENTATION

Know your errors, then reduce them

Systematic error

A consistent error from a faulty instrument or wrong method — e.g. a zero error. It shifts every reading the same way.

Random error

Unpredictable scatter from changing conditions or judgement — e.g. parallax. It varies from reading to reading.

HOW TO MINIMISE ERRORS

- ✓ Take repeated readings and use the average value.
- ✓ Check the zero error of the instrument before use.
- ✓ Avoid parallax — read scales straight on, eye level.
- ✓ Use a finer instrument with a smaller least count.

SUMMARY

Key takeaways

- Measurement compares a quantity with a standard unit.
- The SI system has seven base units.
- Least count sets an instrument's precision.
- A pendulum's period depends on length and gravity.
- Averaging readings reduces error.

QUICK QUIZ • CHECK YOURSELF

1. The SI unit of mass is:

gram

✓ kilogram

newton

2. The least count of a vernier caliper is usually:

1 mm

✓ 0.1 mm

0.01 mm

3. The time period of a simple pendulum does NOT depend on:

length

✓ mass of bob

gravity

Correct answers are marked with

